8.033 Fall 2025: Solution Set 6

1

(a) Since $T_{(ab)}$ is constructed from elements of T_{ab} only, the options are limited. After some trial and errors, we come up with

$$T_{(ab)} = \frac{1}{2}(T_{ab} + T_{ba}). \tag{1}$$

It is easy to check that Eq. (1) indeed satisfies the definition of tensor symmetrization. First, $T_{(ab)}$ needs to be symmetric

$$T_{(ab)} = \frac{1}{2}(T_{ab} + T_{ba}) = \frac{1}{2}(T_{ba} + T_{ab}) = T_{(ba)}. \tag{2}$$

In addition, if T_{ab} is already symmetric ($T_{ab} = T_{ba}$), then

$$T_{(ab)} = \frac{1}{2}(T_{ab} + T_{ba}) = \frac{1}{2}(T_{ab} + T_{ab}) = T_{ab}. \tag{3}$$

(b) We can check that

$$T_{[ab]} = \frac{1}{2}(T_{ab} - T_{ba}) \tag{4}$$

is the antisymmetrization of T_{ab}. First, check that $T_{[ab]}$ is antisymmetric

$$T_{[ba]} = \frac{1}{2}(T_{ba} - T_{ab}) = -T_{[ab]}. \tag{5}$$

In addition, if $T_{ab} = -T_{ba}$, then

$$T_{[ab]} = \frac{1}{2}(T_{ab} - T_{ba}) = \frac{1}{2}(T_{ab} + T_{ab}) = T_{ab}. \tag{6}$$

(c) We can simply add Eq. (1) and Eq. (4) together and get

$$T_{(ab)} + T_{[ab]} = T_{ab}, \tag{7}$$

which is to say that an arbitrary tensor T_{ab} can be written as the sum of a symmetric tensor $T_{(ab)}$ and an antisymmetric tensor $T_{[ab]}$.

(d) We first assume that $T_{ab} = -T_{ba}$ (which is equivalent to $T_{[ab]} = T_{ab}$), then we raise its indices using the metric

$$T^{ab} = g^{aa'} g^{bb'} T_{a'b'} = -g^{aa'} g^{bb'} T_{b'a'} = -T^{ba}. \tag{8}$$

Now we check the other way around. Assuming $T^{ab} = -T^{ba}$, we lower the indices of T^{ab}, we get

$$T_{ab} = g_{aa'} g_{bb'} T^{a'b'} = -g_{aa'} g_{bb'} T^{b'a'} = -T_{ba}. \tag{9}$$

So we have shown that raising and lowering indices preserving antisymmetry of the index.

The same argument applies to the symmetry of the index. We assume that $T_{ab} = T_{ba}$, then

$$T^{ab} = g^{aa'} g^{bb'} T_{a'b'} = g^{aa'} g^{bb'} T_{b'a'} = T^{ba}. \tag{10}$$

Assuming $T^{ab} = T^{ba}$, then

$$T_{ab} = g_{aa'} g_{bb'} T^{a'b'} = g_{aa'} g_{bb'} T^{b'a'} = T_{ba}. \tag{11}$$

So we have shown that raising and lowering indices preserving symmetry of the index.

2

Suppose Alice is boosted with 3-velocity $\vec{v}_{AB}$ in Bob's frame, which we assume not just along the $+x$ direction, and Charlie is boosted with speed v_{CB} along the $+x$ direction in Bob's frame. We are interested in knowing Alice's 3-velocity measured by Charlie in the $+y$ and $+z$ direction, namely v^y_{AC} and v^z_{AC}.

By definition, we have

$$v^y_{AC} = \frac{dy_{AC}}{dt_{AC}}, \tag{12}$$

where y_{AC} is Alice's y coordinate measured by Charlie and t_{AC} is Alice's time coordinate measured by Charlie. To find out how 3-velocities add, we need to express y_{AC} and t_{AC} in terms of quantities measured in Bob's frame. By Lorentz transform, we have

$$y_{AC} = y_{AB}, \tag{13}$$

$$t_{AC} = \gamma_{BC}(t_{AB} + v_{BC} x_{AB}). \tag{14}$$

Plugging into Eq. (12), we get

$$v^y_{AC} = \frac{dy_{AC}}{dt_{AC}} = \frac{dy_{AB}}{d[\gamma_{BC}(t_{AB} + v_{BC} x_{AB})]} = \frac{dy_{AB}}{\gamma_{BC}(dt_{AB} + v_{BC} dx_{AB})} = \frac{v^y_{AB}}{\gamma_{BC}(1 + v_{BC} v^x_{AB})}, \tag{15}$$

where to get the last equality, we have divided the numerator and denominator by dt_{AB} and used the definition

$$v^y_{AB} = \frac{dy_{AB}}{dt_{AB}}, \; v^x_{AB} = \frac{dx_{AB}}{dt_{AB}}. \tag{16}$$

The same logic applies to Alice's 3-velocity measured by Charlie in the z direction

$$v^z_{AC} = \frac{dz_{AC}}{dt_{AC}} = \frac{dz_{AB}}{d[\gamma_{BC}(t_{AB} + v_{BC} x_{AB})]} = \frac{dz_{AB}}{\gamma_{BC}(dt_{AB} + v_{BC} dx_{AB})} = \frac{v^z_{AB}}{\gamma_{BC}(1 + v_{BC} v^x_{AB})}. \tag{17}$$

3

We can choose the axes of the coordinate system such that $\vec{u}$ is aligned with the x axis in frame R. In this reference frame, the particle U has 3-velocity

$$V^x_{UR} = u^x,\ V^y_{UR} = 0, \tag{18}$$

and particle V has 3-velocity

$$V^x_{VR} = v^x,\ V^y_{VR} = v^y, \tag{19}$$

To find out particle V's 3-velocty in V's frame, we just need to apply the 3-velocity addition formula

$$V^x_{VU} = \frac{V^x_{VR} + V^x_{RU}}{1 + V^x_{VR}V^x_{RU}} = \frac{v^x - u^x}{1 - v^x u^x}, \tag{20}$$

and

$$V^y_{VU} = \frac{V^y_{VR}}{\gamma_{RU}(1 + V^x_{RU}V^x_{VR})} = \frac{v^y}{\gamma(|\vec{u}|)(1 - u^x v^x)}. \tag{21}$$

4

(a) We first write $\epsilon_{ijk}\epsilon^{imn}$ as the sum of three terms

$$\epsilon_{ijk}\epsilon^{imn} = \epsilon_{1jk}\epsilon^{1mn} + \epsilon_{2jk}\epsilon^{2mn} + \epsilon_{3jk}\epsilon^{3mn}. \tag{30}$$

Notice that out of the three terms on the right-hand side, at most one of them is non-zero. For example, suppose the first term $\epsilon_{1jk}\epsilon^{1mn}$ is none-zero, which can only be true if one of j, k is 2 and the other is 3 while, at the same time, one of m, n is 2 and the other is 3. So, there are 4 possible combinations of values for j, k, m, and n that would allow $\epsilon_{1jk}\epsilon^{1mn}$ to be none-zero:

$$\begin{aligned}
&j = 2, k = 3, m = 2, n = 3, \text{ so } \epsilon_{ijk}\epsilon^{imn} = \epsilon_{123}\epsilon^{123} = 1,\\
&j = 2, k = 3, m = 3, n = 2, \text{ so } \epsilon_{ijk}\epsilon^{imn} = \epsilon_{123}\epsilon^{132} = -1,\\
&j = 3, k = 2, m = 2, n = 3, \text{ so } \epsilon_{ijk}\epsilon^{imn} = \epsilon_{132}\epsilon^{123} = -1,\\
&j = 3, k = 2, m = 3, n = 2, \text{ so } \epsilon_{ijk}\epsilon^{imn} = \epsilon_{132}\epsilon^{132} = 1.
\end{aligned} \tag{31}$$

Both $\epsilon_{2jk}\epsilon^{2mn}$ and $\epsilon_{3jk}\epsilon^{3mn}$ will be zero in all these four cases due to repeated indices. We can draw essentially the same conclusions if $\epsilon_{2jk}\epsilon^{2mn}$ is the none-zero term or if $\epsilon_{3jk}\epsilon^{3mn}$ is the none-zero term.

From Eq. (31), we see that $\epsilon_{ijk}\epsilon^{imn}$ is 1 only if $j = m$ and $k = n$ while $j \neq k$, and $\epsilon_{ijk}\epsilon^{imn}$ is -1 only if $j = n$ and $k = m$ while $j \neq k$. We can summarize this conclusion with

$$\epsilon_{ijk}\epsilon^{imn} = \delta^m_j\delta^n_k - \delta^n_j\delta^m_k. \tag{32}$$

(b) By the definition of the cross product, we have

$$\begin{aligned}(\vec{A}\times\vec{B})_1 &= A_2B_3 - A_3B_2\\ (\vec{A}\times\vec{B})_2 &= A_3B_1 - A_1B_3\\ (\vec{A}\times\vec{B})_3 &= A_1B_2 - A_2B_1.\end{aligned} \tag{33}$$

We see that this is the same as

$$(\vec{A}\times\vec{B})_i = \epsilon_{ijk}A^jB^k, \tag{34}$$

or equivalently,

$$\vec{A}\times\vec{B} = \epsilon^i{}_{jk}A^jB^k\hat{e}_i. \tag{35}$$

Then, we can write

$$\vec{\nabla}\times\vec{f} = \epsilon^{ij}{}_k\partial_j f^k\hat{e}_i. \tag{36}$$

An additional note: in this problem I am writing some indices in the upper position and some in the lower position just to keep the convention that repeated indices, with one at the upper position and one at the lower position, are summed over. Since we are working in 3-D Euclidean space, an index appearing in the upper position or lower position does not make a physical difference.

5

For any 4 by 4 matrix L, a theorem from linear algebra gives

$$\epsilon_{\alpha\beta\gamma\delta}L^\alpha{}_\mu L^\beta{}_\nu L^\gamma{}_\rho L^\delta{}_\sigma = \epsilon_{\mu\nu\rho\sigma}|L|, \tag{37}$$

where $|L|$ is the determinant of the matrix L.

Now let $L^\mu{}_{\mu'} = \partial x^\mu/\partial x^{\mu'}$, the coordinate transformation matrix, then we get

$$\epsilon_{\alpha\beta\gamma\delta}L^\alpha{}_{\alpha'}L^\beta{}_{\beta'}L^\gamma{}_{\gamma'}L^\delta{}_{\delta'} = \epsilon_{\alpha'\beta'\gamma'\delta'}|L|. \tag{38}$$

The extra factor $|L| = |\partial x^\mu/\partial x^{\mu'}|$ shows that the Levi Civita symbol does not transform like a tensor, so it is not a tensor.

Equivalently, we can write Eq. (38) as

$$\begin{aligned}\epsilon_{\alpha'\beta'\gamma'\delta'} &= |L|^{-1}\epsilon_{\alpha\beta\gamma\delta}L^\alpha{}_{\alpha'}L^\beta{}_{\beta'}L^\gamma{}_{\gamma'}L^\delta{}_{\delta'}\\ &= |L^{-1}|\epsilon_{\alpha\beta\gamma\delta}L^\alpha{}_{\alpha'}L^\beta{}_{\beta'}L^\gamma{}_{\gamma'}L^\delta{}_{\delta'}\\ &= \left|\frac{\partial x^{\mu'}}{\partial x^\mu}\right|\epsilon_{\alpha\beta\gamma\delta}L^\alpha{}_{\alpha'}L^\beta{}_{\beta'}L^\gamma{}_{\gamma'}L^\delta{}_{\delta'}.\end{aligned} \tag{39}$$

We call a quantity that transforms like a tensor but with an extra factor of $\left|\frac{\partial x^{\mu'}}{\partial x^\mu}\right|^n$ a tensor density of weight n. The Levi Civita symbol $\epsilon_{\alpha\beta\gamma\delta}$ is therefore a tensor density of weight 1.

To see how to convert it to a tensor, consider how the metric transforms between coordinates

$$g_{\alpha'\beta'} = \frac{\partial x^\alpha}{\partial x^{\alpha'}} \frac{\partial x^\beta}{\partial x^{\beta'}} g_{\alpha\beta}. \tag{40}$$

Taking the determinant of Eq. (40), we get

$$\det(g_{\alpha'\beta'}) = \left| \frac{\partial x^{\mu'}}{\partial x^\mu} \right|^{-2} \det(g_{\alpha\beta}), \tag{41}$$

where $\det(g_{\alpha\beta})$ is computed if we were to place elements of g_{ab} into a matrix and compute the matrix's determinant. We conclude that the determinant of the metric $\det(g_{\alpha\beta})$ is a scalar tensor density of weight -2. Define $g \equiv \det(g_{\alpha\beta})$ and $g' \equiv \det(g_{\alpha'\beta'})$, and take the absolute value of Eq. (41) and take the square root, we get

$$\sqrt{|g'|} = \left| \frac{\partial x^{\mu'}}{\partial x^\mu} \right|^{-1} \sqrt{|g|}. \tag{42}$$

Therefore, we conclude that $\sqrt{|g|}$ is a scalar tensor density of weight -1. In other words, it transforms with an extra factor of $\left|\frac{\partial x^{\mu'}}{\partial x^\mu}\right|^{-1}$, which would exactly cancel out the extra factor when $\epsilon_{\alpha\beta\gamma\delta}$ transforms. We thus conclude that

$$\tilde{\epsilon}_{\alpha\beta\gamma\delta} = \sqrt{|g|}\epsilon_{\alpha\beta\gamma\delta} \tag{43}$$

is a tensor. More generally, we can always convert a tensor density of weight w into a tensor by multiplying $|g|^{w/2}$.

6

(a) First, we want to check that $V^a_\perp$ is orthogonal to U_a:

$$\begin{aligned} V^a_\perp U_a &= P^a{}_b V^b U_a = (g^a{}_b + U^a U_b) V^b U_a \\ &= (g^a{}_b U_a + U^a U_a U_b) V^b \\ &= (g_{ab} U^a - U_b) V^b \\ &= (U_b - U_b) V^b = 0, \end{aligned} \tag{44}$$

where in the second last line we have used $U^a U_a = -1$ for the 4-velocity vector U^a.

Now we want to check that $V^a_\perp$ is unaffected by further projections.

$$\begin{aligned} P^a{}_b V^b_\perp &= (g^a{}_b + U^a U_b)(g^b{}_c + U^b U_c) V^c \\ &= (g^a{}_b g^b{}_c + U^a U_b g^b{}_c + g^a{}_b U^b U_c + U^a U_b U^b U_c) V^c \\ &= (g^a{}_c + U^a U^b g_{bc} + g^{ab} U_b U_c - U^a U_c) V^c \\ &= (g^a{}_c + U^a U_c + U^a U_c - U^a U_c) V^c \\ &= (g^a{}_c + U^a U_c) V^c = P^a{}_c V^c = V^a_\perp. \end{aligned} \tag{45}$$

(b) From (a), we saw that the definition $P^a{}_b = g^a{}_b + U^a U_b$ worked because the property of 4-velocity that $U^a U_a = -1$. Now suppose we have an arbitrary timelike 4-vector W^a, in analogy, we can define the projection operator as $P^a{}_b = g^a{}_b + nW^a W_b$, where n is a scalar taken into account that $W^a W_a$ is no longer -1. Similarly to how the term $U^a U_b$ works for a 4-velocity, we want $nW^a W_b W_a = -W_b$, so we just need to set $n = -1/(W^a W_a)$. Therefore, in this case the projection tensor is

$$P^a{}_b = g^a{}_b - \frac{W^a W_b}{W^c W_c}. \tag{46}$$

Comment: Note that I did not write

$$P^a{}_b = g^a{}_b - \frac{W^a W_b}{W^a W_a}$$

because the index a would appear three times with the one in the numerator being a free index and the two in the denominator being dummy indices, which is not good!

(c) Although we assumed W^a was an arbitrary *timelike* vector, our definition o f the projection operator in Eq. (46) did not rely on whether $W^a W_a < 0$ or > 0. So the same definition in Eq. (46) can be applied to an arbitrary spacelike vector as well!

7

(a) To check if $V^a + W^a$ is spacelike, we just need to check if $(V^a + W^a)(V_a + W_a) > 0$.

$$(V^a + W^a)(V_a + W_a) = V^a V_a + 2V^a W_a + W^a W_a. \tag{47}$$

We are given that V^a and W^a are spacelike, so $V^a V_a > 0$ and $W^a W_a > 0$. In addition, it is given that $V^a W_a = 0$, so we conclude that

$$(V^a + W^a)(V_a + W_a) = V^a V_a + W^a W_a > 0, \tag{48}$$

so we have shown that $V^a + W^a$ is spacelike.

(b) k^a is a null vector, so we can choose our coordinates such that $k^1 = k^0, k^1 = k^2 = 0$, where $k^0 \neq 0$. In this frame,

$$U^a k_a = -U^0 k^0 + U^1 k^1 = -U^0 k^0 + U^1 k^0 = k^0(U^1 - U^0). \tag{49}$$

Since k^a is a null vector, $k^0 \neq 0$. We just need to show that $U^1 - U^0 \neq 0$, which we will show by contradiction. Suppose $U^1 - U^0 = 0$ (or equivalently $U^0 = U^1$), then the norm of U^a is

$$U^a U_a = -(U^0)^2 + (U^1)^2 + (U^2)^2 + (U^3)^2 = (U^2)^2 + (U^3)^2. \tag{50}$$

We are given that U^a is a timelike vector, so U^2 and U^3 cannot be both zero, which would otherwise indicate U^a is a null vector. Then $U^a U_a = (U^2)^2 + (U^3)^2 > 0$ which contradicts that U^a is a timelike vector. We thus conclude that $U^0 \neq U^1$, so Eq. (49) cannot be zero. Note that although we did this proof by choosing a specific frame such that $k^1 = k^2 = 0$, $U^a k_a$ is a Lorentz scalar so it is zero in all frames. We are done with the proof.

(c) The statement is false. We can pick an arbitrary 4-velocity U^a and pick $V^a = U^a$. We see that $U^a - V^a = 0$, but 4-velocities must have their norms equal to -1, so we conclude that $U^a - V^a$ is not a 4-velocity in general!

8

t^a and k^a being future-directed means $t^0 > 0$ and $k^0 > 0$ in any reference frame (given that they are both timelike vectors). Then it is simple to verify that $v^a = t^a + k^a$ is also future-directed. Since $v^0 = t^0 + k^0$, the sum of two positive numbers guarantees that $v^0 > 0$.

To show that $v^a = t^a + k^a$ is timelike, we can choose an inertial reference frame where $k^1 = k^0, k^2 = k^3 = 0$. Then

$$v^a v_a = (t^a + k^a)(t_a + k_a) = t^a t_a + 2t^a k_a + k^a k_a = t^a t_a + 2t^a k_a. \tag{51}$$

Since t^a is timelike, $t^a t_a < 0$. We just need to check if $t^a k_a < 0$. We write

$$t^a k_a = t^0 k_0 + t^1 k_1 = -t^0 k^0 + t^1 k^0 = k^0(t^1 - t^0). \tag{52}$$

Since k^a is future directed, $k^0 > 0$. t^a being timelike implies that

$$(t^0)^2 > (t^1)^2 + (t^2)^2 + (t^3)^2 > (t^1)^2. \tag{53}$$

Combining Eq. (53) with $t^0 > 0$, we conclude that $t^0 > t^1$, so $t^1 - t^0 < 0$. Then $t^a k_a = k^0(t^1 - t^0) < 0$. We have thus shown that $v^a = t^a + k^a$ is timelike.